

FOX_Checklist

What this is: every FoxByte feature and its sub-features — built or not, and how it was tested — so it's clear where we are and what's next. How each feature works is in `FOX_Feature_Implemented.pdf` (same ids).

Last updated: 22 September 2026 · **Reflects:** `main @ 2778762` (v1.0.1, exact repository path PR #1) + the FoxByte rename and brand indirection (`chore/foxbyte`)

Source: `docs/FOX_Checklist.html` → `make feature-doc` → this PDF

☒ Done & tested — automated tests (unit, VM integration or CI)

☒ Done — tested by hand only

☐ Partial — gaps, or not verified end to end

☒ Known issue

☐ Not started

Sub-features: ☒ built ☐ partly ☐ not built yet ☒ broken · "VM §n" = section of `make integration` ; "v2 §n" = `make integration-v2` ; "update" = `make integration-update`

Where we are

Area	Features	Done & tested	Done (manual)	Partial	Known issue	Not started	Progress
A · Install & setup	9	4	3	2	·	·	78%
B · CLI & services	2	2	·	·	·	·	100%
C · Branching & storage	8	7	·	1	·	·	88%
D · Durability	6	5	·	1	·	·	83%
E · Security foundations	3	3	·	·	·	·	100%
F · SQL gateway	7	6	1	·	·	·	100%
G · Accounts & auth	6	5	·	1	·	·	83%
H · Blackbox (record of schema changes)	18	17	·	1	·	·	94%
I · Import & migration	13	5	8	·	·	·	100%
J · ETL pipelines (experimental)	4	2	2	·	·	·	100%
K · APIs, SDKs & agents	6	4	2	·	·	·	100%
L · Web console	11	1	10	·	·	·	100%
M · Build, CI & releases	7	6	1	·	·	·	100%
Total (features built or started)	100	67	27	6	·	·	94%
N · Not started (roadmap)	11	1	·	·	·	10	9%

Storage engine — not implemented, by design

FoxByte does **not** have its own storage engine. Data is stored by **PostgreSQL's own engine** (heap tables, B-tree indexes, write-ahead log) running on local disk inside the stock `postgres:18-bookworm` image plus `wal-g` (`postgres:16-bookworm` for an install created before 21 Sep 2026). This was a locked architecture decision: a Neon-style engine that separates compute from storage over the network adds latency to every transaction, and transaction speed must never be compromised.

What **is** built is a storage *layer* around Postgres: copy-on-write branch drivers — ZFS on macOS/Linux, btrfs on Windows — behind one interface in `internal/branch/storage.go` (C1), quotas (C6), capacity reporting (C7), and durability to object storage with `wal-g` + MinIO: WAL archiving, base backups, point-in-time restore (D1–D3). Networked storage disaggregation is listed in the README roadmap but not started (see "Not started").

Suggested next steps

- Security hardening from the 22 Sep audit** — done but for the last part: the VM suites nightly in CI, and UI (Playwright) and SDK contract tests (G24, G26).
- Cut the first signed release** and check `gh attestation verify` and `fox update` against it (M4, A9).

3. Run on real machines what the throwaway VM cannot: `fox update` across a real release on macOS and Windows (A9); on Windows, the `zstd` distro image's unpack (M5), the `btrfs` holding area of `fox pg upgrade` (D6), and an export's copy to the host (D5).
4. Make the `postgres-wal-g` image package public, so installs pull it instead of building it for a few minutes (C8).
5. Run the VM integration suites in CI on every pull request (M2).
6. Add automated tests where there are none: imports (I1–I3, I7), SDKs (K3, K4), web UI (L*).
7. HA beyond the demo: automatic failover and fencing (D4).

ID	Feature · sub-features	Tested by
 A1	One-line installer (macOS / Linux) ✓ Detects OS / CPU, downloads the release binary ✓ Installs to the prefix (sudo only if needed) ✓ Stages the Linux engine for the VM on macOS	Unit (<code>test_install_sh.sh</code> , CI); manual
 A2	Windows installer (install.ps1) ✓ Installs WSL, elevates, resumes after reboot ✓ Downloads fox.exe, engine, image build context ✓ Ubuntu rootfs verified against Ubuntu's checksums ✓ Adds to PATH and runs <code>fox setup</code>	Pester (Windows CI); Windows run 16 Sep
 A3	fox setup on macOS (Lima) ✓ Creates / starts the Lima VM ✓ Installs Docker and ZFS ✓ Reinstalls the engine on every run ✓ Starts the stack	Unit; run against a deliberately broken VM, 21 Sep
 A4	fox setup on Windows (WSL2 + btrfs) ✓ Imports a dedicated distro with systemd ✓ Docker, btrfs and a storage mount unit ✓ Quiet steps, detail in install.log ✓ Waits for systemd on a cold distro	Unit (<code>host_wsl_test.go</code>); Windows run 16 Sep
 A5	Linux native ✓ Engine runs in-process, no VM ✓ <code>fox setup</code> explains nothing is needed	VM integration (all suites run on Linux)
 A6	Command forwarding & host file streaming ✓ macOS → Lima VM, starts the VM if stopped ✓ Windows → WSL distro	Unit (<code>TestImportLocalFile</code> , <code>TestWSLargs</code>)
 A7	Always-latest engine binary ✓ Downloads the newest engine into the cache (ELF check) ✓ <code>FOX_NO_REFRESH</code> , <code>FOX_VERSION</code>	Unit (<code>host_test.go</code> , <code>update_test.go</code>); Windows run 16 Sep
 A8	New-release notice on fox start ✓ Background check, 1.5 s limit, printed last ✓ Only a promoted, newer, complete release	Unit; update §1, §1b, §2, §3, §8; real Mac check
 A9	fox update ✓ Downloads and verifies everything before changing anything ✓ Stops servers, installs engine, restarts old one on failure ✓ <code>up</code> , Blackbox upgrade, <code>start</code> on the new engine ✓ Replaces host <code>fox</code> , keeps previous binaries	Unit; update §4–§7; a real Mac run

ID	Feature · sub-features	Tested by
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✓	B1 The fox command set	✓ Stack: setup, start, stop, up, down, status, logs, psql, update ✓ Branching, backup / restore, HA ✓ Blackbox, policy, impact ✓ Import, pipelines ✓ backup export / restore, pg status / upgrade	✓ Users, API keys, admin ✓ <code>fox vm status / fox vm shell</code> (Lima, WSL; Linux says there is no VM) ✓ <code>branch create --from</code> ✓ <code>fox uninstall</code> removes it all, retired installs included	Unit (<code>main_test.go</code> , <code>update_test.go</code>); VM suites
✓	B2 Background services (daemon)	✓ Control plane :8080, gateway :6432, agent API :8088 ✓ Detached, pidfiles and logs in <code>~/ .fox</code>	✓ Stop (TERM then KILL), status, logs □ Health check / restart on crash	Unit (<code>TestStopTerminatesProcess</code>); VM §1

C Branching & storage

7 of 8 done

ID	Feature · sub-features	Tested by
✓	C1 Storage drivers (ZFS / btrfs) ✓ ZFS snapshot + clone per branch (macOS VM, Linux) ✓ btrfs subvolume snapshots (Windows) ✓ Pool / filesystem created on a loopback file	Unit; VM §2, §2b (ZFS); btrfs manual on Windows
✓	C2 Branch lifecycle ✓ Create, list, delete ✓ Reset to parent ✓ Suspend / resume ✓ Recreate a container from its surviving data	Unit; VM §2, §2b, §4
✓	C3 Protection of main ✓ Delete and reset refused	Unit; VM §4, §6
✓	C4 Private networking (no host ports) ✓ Branch databases reachable only through the gateway ✓ Published ports (object store, restore, debug) on loopback only; standby unpublished	Unit (<code>TestParsePublishedPort</code> , <code>TestDSN</code>); VM §1c
✓	C5 Scale-to-zero & wake on connect ✓ Idle branches suspended automatically ✓ Next connection wakes the branch (~0.3–0.7 s)	VM §7
🔥	C6 Storage quotas ✓ Space reserved for main (ZFS) 🔥 Per-branch cap (opt-in)	Manual
✓	C7 Capacity & status ✓ Services, main health, backups, branches ✓ Storage used / available	VM §1
✓	C8 Provisioning & image pull ✓ Docker, storage pool and image prepared on up	Unit; VM §11b
		✓ Pulls from GHCR, builds locally as fallback ✓ The build context is built into fox, so the fallback works on any install

D Durability

5 of 6 done

ID	Feature · sub-features	Tested by
✓	D1 WAL archiving to object storage ✓ wal-g ships main's WAL to MinIO, off the commit path ✓ MinIO images pinned on quay.io ✓ Branches are disposable by design (export one that matters)	VM §3; VM §11c
		✓ Remote S3 target (<code>fox backup target set</code>), keys in 0600 env files ✓ Object Lock reported; tested on MinIO □ Tested against AWS S3

✓	D2 Base backups	✓ Create and list ✓ Follow the primary after a failover; listing works with main stopped ✓ Read-only listing: <code>GET /api/backups</code> and a Dashboard panel	✓ Schedule and retention (<code>FOX_BACKUP_INTERVAL</code> , <code>FOX_BACKUP_RETAIN</code>) ✓ A first backup at install ✓ Age shown and warned in <code>fox status</code> and the status API □ Automated test restore	VM §3, §3b, §6; VM §11c
✓	D3 Point-in-time restore	✓ Restore to a timestamp or latest on :5433 ✓ Per-install credentials ✓ Starts from the newest base backup that precedes the target	✓ A point no backup precedes is refused, naming the oldest	Unit; VM §3, §3b, §6; v2 §1a
🔥	D4 HA standby & failover	✓ Streaming hot standby ✓ Manual failover, <code>main</code> rerouted ✓ Nothing deletes or stops the promoted standby ✓ <code>fox up</code> brings it back, not the old main	✓ <code>fox ha fallback</code>, keeping every write ✓ The promoted standby archives WAL, so backups and PITR cover the failover □ Automatic failover, fencing, multi-host	Unit; VM §6, §7
✓	D5 Portable export & restore	✓ Every non-agent branch: <code>pg_dump</code>, its roles, the Blackbox head; SHA256SUMS over all ✓ Copied out of the VM and verified on the host; a copy that fails is deleted ✓ Damaged, incomplete or path-escaping archives refused	✓ Restore into a new branch, same or newer major; Blackbox checked exact 🔥 Host copy on macOS/Windows unit-tested only	Unit; VM pg-upgrade §1
✓	D6 PostgreSQL major upgrade	✓ <code>fox pg status</code> : the major of the install and of each branch ✓ Plan with every change crossed, found-in-your-data first; refusals say what to do ✓ Asks, exports, then reloads every carried branch on the new major	✓ Data, roles, API keys and every Blackbox entry come across ✓ Old databases held outside the branch namespace; rollback and finalize ✓ Each upgraded cluster archives WAL under its own prefix 🔥 btrfs (Windows) holding area not yet run □ Carry agent branches; upgrade with HA on	Unit; VM pg-upgrade (72 / 72)

E Security foundations 3 of 3 done

ID	Feature · sub-features	Tested by
✓	E1 Per-install generated secrets ✓ Generated Postgres and MinIO passwords ✓ Used by restore too ✓ Each login role has its own derived password	Unit (indirect); v2 §1a; VM §1d
✓	E2 TLS everywhere ✓ Self-signed TLS on all three services, zero setup	Unit (<code>TestReadStartupUpgradesTLS</code>)
✓	E3 Keys are made by a person, not by start ✓ Start creates no account and no key, and caches none on disk ✓ The banner shows the connection string and how to make a key ✓ The first account created gets the admin grant on main	Integration §1

F SQL gateway 7 of 7 done

ID	Feature · sub-features	Tested by
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✓	F1 Routing by database name	✓ One endpoint :6432, database name = branch □ Connection pooling, rate limiting	Unit; VM §1, §6
✓	F2 API-key authentication	✓ key_ key as the Postgres password ✓ Startup deadline and connection cap ✓ Bad key rejected with a Postgres error ✓ An account key opens main, its own branches, or any for an admin	VM §1
✓	F3 Backend login (SCRAM)	✓ Cleartext, md5, SCRAM-SHA-256 □ Unit test for SCRAM ✓ Clients never see the database password	VM §1, §6, §8 (exercised)
✓	F4 Per-user Postgres login roles	✓ Role per user, actor can't be forged ✓ Own password per role ✓ Permissions of non-superuser db_client	Manual
✓	F5 Attribution session settings	✓ Actor kind, branch, session id per connection ○ Kind / branch / session client-settable	VM §8
✓	F6 Ctrl-C (CancelRequest) forwarding	✓ Cancel a running query through the gateway	Unit
✓	F7 Errors & insecure build tag	✓ Real Postgres error responses ✓ No-auth bypass impossible in release builds	Unit (TestSendError)

G Accounts & auth 5 of 6 done

ID	Feature · sub-features	Tested by
✓	G1 Auth store (SQLite) ✓ Users, keys, sessions, OAuth ids, pipelines ✓ Safe under concurrent use (WAL, busy timeout), including a new store created by three services at once □ Roles / permissions, multi-tenant projects	Unit (TestVerifyKeyConcurrent , TestConcurrentFirstOpen); VM §1
✓	G2 Email & password users ✓ Create account, log in ✓ Password change (signs out other sessions); admin reset at the CLI □ Self-service reset by email ✓ List and delete users (admins) ✓ Emails validated; at most 63 bytes (a role name)	Unit; VM §1; VM §1d
✓	G3 API keys ✓ Create (shown once), list, revoke ○ Revoking an unknown id reports success ✓ Bearer / X-API-Key and gateway password	Unit; VM §1
✓	G4 Web sessions & signup control ✓ httpOnly session cookie ✓ Failed sign-ins throttled ✓ Sign-up closed by default (FOX_SIGNUP=open)	Unit (TestSessions)
○	G5 OAuth (GitHub / Google) ✓ Sign-in flows built □ Tested with a real OAuth app ✓ Defaults match what the engine serves (https://localhost:8080 ; web origin = public URL) ✓ Only a provider-verified email joins or creates an account; closed sign-up respected ✓ Session cookie Secure / SameSite follow the origin ✓ State cookie one-use, Secure , SameSite=Lax	Unit; VM §1b
✓	G6 Security log ✓ Sign-ins, failures, keys, passwords, accounts, admin grants, refusals ✓ fox_audit , verify , GET /api/audit ✓ Hash-chained, append-only, anchored and signed □ Included in fox backup export	Unit; VM §1d

H Blackbox (record of schema changes) 17 of 18 done

ID	Feature · sub-features	Tested by
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✓	H1 DDL recording	✓ Who, tool, branch, statement, status, risk ✓ Inherited by every branch ✓ A restart records nothing: the install script's own statements are not captured ❑ Role / database changes, some dropped object types	✓ Agents' UPDATE and DELETE recorded; TRUNCATE by anyone ✓ Triggers on every table, protected from drop/disable/rename	VM §8; v2 §9; v2 §10
🔗	H2 Hash chain, append-only & verify	✓ Rows can't be edited or deleted normally ✓ <code>fox ledger verify</code> 🔗 Superuser rewrite caught only via anchors (H10–H12)	✓ <code>bb.policy</code> changes recorded (append-only, not chained)	VM §8; v2 §0, §6b
✓	H3 Attribution	✓ Humans via their login role ✓ Agents via their own <code>agent-<id></code> role	✓ Kind derived from the role, so neither can pose as the other ✓ Tool and session are client-declared by design, and labelled so	Unit; VM §8; v2 §1c
✓	H4 Destructive-DDL guardrail	✓ DROP TABLE / DROP SCHEMA refused, attempt recorded ✓ Override only superusers / <code>db_admin</code>, use recorded	✓ "flag" and "allow" actions; unknown actions refused ✓ No hang when blocked after a write in the same transaction ✓ TRUNCATE blocked by default, recorded	VM §8; v2 §1b, §6b
✓	H5 Risk flags	✓ Type change, drop column flagged ✓ Judged on the running statement (no comments or literals)	✓ Automated test	Unit; v2 §6b
✓	H6 Blackbox access (CLI / REST / MCP / UI / SDK)	✓ CLI, REST with filters, MCP, web page, SDKs	❑ Per-user access control	v2 §0, §5
✓	H7 <code>fox ledger revert --to</code>	✓ Help and output say it's a point-in-time restore of main	✓ Points to H13 for a branch from before a change	Unit (help text); VM §3 (the restore)
✓	H8 Override permission (<code>db_admin</code>) & upgrade	✓ <code>fox admin grant revoke list</code> ✓ First user granted automatically	✓ <code>fox blackbox upgrade [--all]</code> ✓ Grant / revoke from the web (admins only)	v2 §1b, §6
✓	H9 Blackbox 2.0 capture	✓ Transaction id and WAL position per entry ✓ Task, tool-call hash, override use	✓ Fail-safe, append-only ✓ Kill switch not settable by clients	Unit; v2 §2
✓	H10 Checkpoints & anchors	✓ Merkle root in the database and in read-only anchor files ✓ On a schedule and on demand ✓ Published anchor format	✓ Copied to the backup target (S3 Object Lock) ✓ Anchors signed (Ed25519, key outside the databases)	Unit; v2 §3, §3b
✓	H11 <code>fox-verify</code> (independent verifier)	✓ Checks a live branch or an export ✓ Release binaries for every platform	✓ <code>--pubkey</code> checks signatures	Unit; v2 §3
✓	H12 Blackbox integrity & export	✓ INTACT / TAMPERED via CLI, REST, MCP ✓ JSON-lines export	❑ Streaming for very large ledgers	v2 §3
✓	H13 Branch from just before a Blackbox entry	✓ Restore + replay up to the change's transaction ✓ By transaction id or timestamp	✓ CLI, REST, MCP, web panel ✓ main untouched	Unit; v2 §4
✓	H14 Blackbox names & aliases	✓ <code>fox blackbox</code>, <code>/blackbox</code> routes, MCP and SDK names	✓ Every original name still works	Unit; v2 §5

✓	H15 Policy gate (warn / block)	✓ BBX02 warn / BBX01 block with JSON detail ✓ Blocked attempts recorded ✓ Per-rule <code>db_admin</code> override	✓ CLI, REST, MCP, web page; custom rules ✓ Patterns match the running statement ✓ TRUNCATE (its own guardrail, H4)	Unit; v2 §6, §6b
✓	H16 Agent provenance & execute_change	✓ Session, task, parent session on every agent change (also through the Gateway DSN) ✓ MCP <code>execute_change</code> with policy preview	✓ <code>fox blackbox sessions</code>, REST ⦿ Provenance is what the agent says (not verified)	Unit; v2 §7
✓	H17 Impact analysis	✓ Dependents (views, foreign keys, indexes, depth) ✓ Other branches that have the object ✓ Policy verdict and risk score	✓ CLI, REST, MCP, inside <code>execute_change</code> ⦿ Functions, types, schemas not recognised from SQL	Unit; v2 §8
✓	H18 Blackbox diff between branches	✓ Fork point, changes only on each side ✓ Objects changed on both	✓ CLI, REST, MCP ☐ Merge (diff only today)	Unit; v2 §8

I Import & migration

13 of 13 done

ID	Feature · sub-features	Tested by
✓ I1	From Postgres ✓ Schema and data via <code>pg_dump</code> ✓ Zero-table result reported as an error	☐ Automated test Manual
✓ I2	From MySQL ≤5.7 / MariaDB ✓ pgloader with type mapping (multi-arch image)	☐ Automated test Manual
✓ I3	From MySQL 8.x ✓ Native <code>mysqldump</code> path (<code>caching_sha2_auth</code>) ✓ Types, NULLs, unicode, PK	☐ Other indexes, foreign keys, defaults ☐ Automated test Manual
✓ I4	From MongoDB ✓ Collection → typed table, exact name	✓ Nested objects / arrays kept as <code>jsonb</code> VM §9, §10
✓ I5	From a .sql dump ✓ Runs the dump into a new branch ✓ Fails listing every failed statement (instance kept)	✓ Same for Postgres imports (I1) ✓ Automated test Unit; VM §11
✓ I6	From JSON / NDJSON ✓ Columns inferred from top-level keys	✓ An invalid document fails the import, naming its element or line Unit; VM §10, §11
✓ I7	From CSV ✓ Table from the header, exact column names	☐ Typed columns (all text today) Manual
✓ I8	Continuous replication & cutover ✓ Initial copy + streaming (logical replication) ✓ Cutover keeps data (CLI) ✓ Status and cutover from REST and the Import page	✓ A replicating branch is never auto-suspended ✓ Automated test (VM §5b) and driven in the browser Unit; VM §5b
✓ I9	Guard handling during import ✓ Guardrail paused, recording stays on	✓ Re-armed even after a crash Manual
✓ I10	Live progress (CLI & SSE) ✓ Live log and progress bar	⦿ <code>psql</code> subprocess output not streamed Manual
✓ I11	Import REST endpoints ✓ From a connection string or an uploaded file ☐ Target name validation	✓ Private, local and container sources for admins only; metadata refused Manual; unit; VM §1d

✓	I12 Web Import page ✓ Database and file panels with live console ✓ No longer claims a SQLite loader that doesn't exist	Manual
✓	I13 Exact identifier fidelity ✓ Case kept (<code>userId</code> , <code>leaseAiChats</code>) ⚠ Field order of JSON sources not kept	Unit; VM §10

J ETL pipelines (experimental) 4 of 4 done

ID	Feature · sub-features	Tested by
✓ J1	Pipeline engine ✓ Extract → raw schema → SQL models → tests, in a fresh branch ✓ <code>ref()</code> / <code>source()</code> templating □ Dependency graph □ Incremental models □ Scheduling ✓ A run replaces only a branch the pipeline made ✓ API sources are connection strings, never server paths	Unit; VM §9
✓ J2	Data-quality tests ✓ <code>not_null</code> , <code>unique</code> , <code>accepted_values</code> , <code>row_count_min</code> , <code>custom</code> ✓ Failure → exit 1, data kept	Unit; VM §9
✓ J3	Stored pipelines & run history ✓ Saved per user, run over the API, history □ Read a run's log □ Mark runs whose process died	Manual
✓ J4	Pipeline editor (web) ✓ Create, edit, run, results, history □ Confirm before delete	Manual

K APIs, SDKs & agents 6 of 6 done

ID	Feature · sub-features	Tested by
✓ K1	Control-plane REST API ✓ Branches, query, Blackbox, policy, impact, import, pipelines, keys □ Versioned paths (<code>/api/v1</code>) ✓ Exact-origin CORS, branch-name check, body limits, timeouts ✓ Branches owned by their maker; others 404 (owner, admin, or main)	Unit; VM §1; v2 §0–§8; VM §1d
✓ K2	OpenAPI spec ✓ Served, Blackbox 2.0 endpoints documented ✓ Matches the code (import fields, status codes, auth routes, <code>/blackbox</code> aliases) ✓ Drift check in CI	Unit (drift check)
✓ K3	Python SDK ✓ Branches, query, Blackbox □ Import, pipelines, keys, agents □ Tests □ Publish to PyPI	Manual
✓ K4	TypeScript SDK ✓ Same client for Node and browsers □ Tests □ Publish to npm	Manual
✓ K5	Agent Branch API ✓ A branch per agent, own non-superuser role ✓ Provenance (task, parent, session) ✓ DSN reachable from the host (gateway-routed, branch-scoped key) ✓ Cap, listing and TTL count suspended branches ✓ Superuser switch only in <code>-tags insecure</code> builds ✓ Agents and their keys belong to the caller	Unit; VM §5; v2 §1c, §7
✓ K6	MCP server ✓ Branches, <code>run_sql</code> , Blackbox tools ✓ <code>policy_check</code> , <code>execute_change</code> , <code>impact</code> , <code>diff</code> ✓ Requires an API key and acts as that account; a scoped key is held to its branch ✓ Client config & 16-tool reference (<code>docs/mcp.md</code>) ✓ Same ownership rule as the REST API	Unit; v2 §1c–§8

L Web console 11 of 11 done

ID	Feature · sub-features	Tested by
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✓	L1 Embedded single-origin UI ✓ Served by the engine at :8080 ☐ Automated UI tests (whole area)	Manual
✓	L2 Login & signup ✓ Log in / sign up tabs, OAuth buttons	Manual
✓	L3 Dashboard ✓ Live tiles ✓ Branch table: create, suspend, resume, delete, copy DSN	Manual
✓	L4 SQL console & browser ✓ Schema sidebar, rows, structure, indexes ✓ Values shown as Postgres writes them (uuid, interval, inet, bytea) ✓ SQL editor ✓ System tables hidden by default, with a show button ✓ Runs as the signed-in user; one-run override switch ✓ Per-rule override for policy blocks	Manual; API in v2 §6
✓	L5 Blackbox page ✓ Entries per branch with badges ✓ Table / risk / date filters, and "Clear filters" ✓ Verify button; a row expands to its statement and session	v2 §5b; browser
✓	L6 API keys page ✓ Create, copy, list, revoke ✓ A new key is shown inside its connection string (E3's next step)	Manual; browser review 21 Sep
✓	L7 Landing, Guide & Docs pages ✓ Landing, developer guide, reference ✓ Refreshed against the code (https, sslmode, MCP, Blackbox, import routes)	Type-check + build
✓	L8 Blackbox integrity page ✓ Verify, checkpoint, export, branch-before ✓ Used signed in; every result matched the CLI ⚠ Entry list needs a refresh to show a new change	Type-check + build; browser review 16 Sep
✓	L9 Policies page ✓ Rules, warn/block, custom rules, preview, evaluations ✓ Used signed in; matched the CLI, wrote only to the chosen branch ✓ Non-admin change refused with a readable message ✓ Who can override: list, grant, revoke	Type-check + build; API in v2 §6; browser review 16 Sep
✓	L10 App sidebar layout ✓ Every link opens its page, highlighted, with breadcrumb ✓ Collapse persists across a reload; theme toggle persists ✓ No console errors on any page ☐ Under-860px drawer (needs a manual look)	Type-check + build; browser review 16 Sep
✓	L11 Design system & FoxByte theme ✓ Tokens for colour, type, space, radius, elevation and motion; components use no raw values ✓ Ember palette in both themes; filled controls legible (--accent-ink) ✓ Fox mark from theme variables, wordmark from brand.json, favicon ✓ Console, Blackbox, Policies and API keys re-laid out; one table style ✓ Every page reviewed in both themes; console checked at 375px ☐ Not yet reviewed against a running engine with real data	Type-check + build; browser review 21 Sep

M

Build, CI & releases

7 of 7 done

ID	Feature · sub-features	Tested by
✓ M1	Makefile targets ✓ build, vet, test, integration suites ✓ Suites run only in a throwaway test VM (guarded) ✓ release, web, WSL builds, feature-doc (both PDFs)	CI
✓ M2	Linux CI ✓ Build, vet, test, gofmt ✓ Release script tests, checklist sync check ☐ VM integration suites in CI ✓ Secret scanning, govulncheck, npm audit, UI type-check	CI

✓	M3	Windows CI	CI
✓		Build, vet, test, Pester, installer encoding	
✓		Guard tests read backslash paths, and generated files are pinned to LF (21 Sep)	
		□ End-to-end WSL install test	
✓	M4	Releases & GHCR image	Script tests in CI
✓		Binaries for 5 platforms + fox-verify	
✓		SHA256SUMS merged safely between jobs	
✓		Fails unless fox update 's files are published	
		✓ Multi-arch Postgres image on GHCR	
		✓ Actions pinned by SHA; signed SHA256SUMS; build attestations; SBOM	
		□ First signed release run on GitHub	
✓	M5	WSL distro & ZFS builds	Actions artifacts; Pester; Windows run 16 Sep
✓		Prebuilt distro image	
✓		Downloaded and verified by the installer (seen on Windows)	
✓		Preloaded image tagged as the engine runs it	
		✓ Imported on a fresh Windows install; preloaded images loaded	
		✓ Published zstd-compressed, with the gzip size measured beside it in every build log	
		🟡 zstd unpack on a real Windows install (unit-tested only); the gzip image was 685 MB	
✓	M6	Postgres image	Unit; all three suites on 18
✓		PostgreSQL 18 + wal-g for a fresh install; one image per supported major (16, 18)	
✓		The image follows the data: an install on 16 keeps running 16 after an update	
✓		A branch on the wrong major is refused before docker run , with the way out	
✓		The major lives in one place, and a test holds the Dockerfile, release matrix and Windows preload to it	
		✓ All three suites pass on PostgreSQL 18.6: 108 + 277 + 51 checks (21 Sep)	
		✓ A command that moves an install from 16 to 18 (D6), and a portable backup to take first (D5)	
		□ pgvector	
		✓ Base image pinned by digest; wal-g checked by SHA-256	
✓	M7	Product name in one file	Unit (brand, gen, oldnames), in CI via go test
✓		brand.json → Go, shell, web and installer blocks, regenerated by make brand	
✓		Stored names brand-free and frozen: a rename never migrates an install	
✓		Retired env prefixes, state directory, VM names and installs still handled	
		✓ Guards: retired names anywhere, product names in stored SQL, durable names spelt twice	
		🟡 The sweep is text substitution: review its diff (it over-reached twice)	

Planned or discussed, nothing built yet. When work starts, move the row into its area with the next id there.

ID	Item · scope	Related
☐	Multi-tenant projects & roles ☐ Projects / organisations ☐ Per-branch permissions (RBAC)	G1, H6
☐	Branch merge ☐ Apply one branch's schema changes to another	H18
☐	Scheduled backups & retention	D2
☐	Automatic HA failover ☐ Health detection, fencing ☐ Multi-host	D4
☐	Pipelines: scheduling, incremental / CDC extract, DAG view	J1
☐	More import sources: Cassandra, DynamoDB	I1–I8
☐	pgvector in the Postgres image	M6
☐	Cloud write-once anchors (S3 Object Lock etc.)	H10
☒	Release signatures — done 22 Sep 2026	A9, M4
☐	Hosted cloud tier	—
☐	Networked storage disaggregation (README roadmap; see the storage engine note on page 1)	C1, D1

Out of scope by decision

A custom storage engine (PostgreSQL's own engine is used, for native transaction speed), sharding, Raft / distributed transactions, multi-region, HTAP.